pumping enough energy into a string, you could even make it grow to macroscopic size. With today's technology we couldn't come anywhere near achieving this, but it's possible that in the searingly hot, extremely energetic aftermath of the big bang, long strings were produced. If some have managed to survive until today, they could very well stretch clear across the sky. Although a long shot, it's even possible that such long strings could leave tiny but detectable imprints on the data we receive from space, perhaps allowing string theory to be confirmed one day through astronomical observations.

Higher-dimensional *p*-branes need not be tiny, either, and because they have more dimensions than strings do, a qualitatively new possibility opens up. When we picture a long—perhaps infinitely long—string, we envision a long one-dimensional object that exists within the three large space dimensions of everyday life. A power line stretched as far as the eye can see provides a reasonable image. Similarly, if we picture a large—perhaps infinitely large—two-brane, we envision a large two-dimensional surface that exists within the three large space dimensions of common experience. I don't know of a realistic analogy, but a ridiculously huge drive-in movie screen, extremely thin but as high and as wide as the eye can see, offers a visual image to latch on to. When it comes to a large three-brane, though, we find ourselves in a qualitatively new situation. A three-brane has three dimensions, so if it were large—perhaps infinitely large—it would *fill* all three big spatial dimensions. Whereas a one-brane and a two-brane, like the power line and movie screen, are objects that exist *within* our three large space dimensions, a large three-brane would occupy all the space of which we're aware.

This raises an intriguing possibility. Might we, right now, be living within a three-brane? Like Snow White, whose world exists within a two-dimensional movie screen—a two-brane—that itself resides within a higher-dimensional universe (the three space dimensions of the movie theater), might everything we know exist within a three-dimensional screen—a three-brane—that itself resides within the higher-dimensional universe of string/M-theory? Could it be that what Newton, Leibniz, Mach, and Einstein called three-dimensional space is actually a particular three-dimensional entity in string/M-theory? Or, in more relativistic language, could it be that the four-dimensional spacetime developed by Minkowski and Einstein is actually the wake of a three-